

Measurement of the Restgas distribution with PANDA

NRW-FAIR Netzwerk

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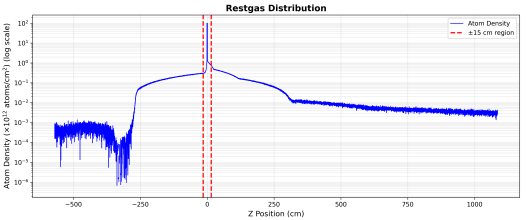
Restgas Distribution & Luminosity Determination

The Restgas Profile

- Simulated from Münster.
- Continuous distribution along the beam pipe.

Previous Achievements

- `LuminosityFit` utilizes Restgas for luminosity determination.
- Proven precision:
  $\Delta L/L < 0.5\%$
  (for $p_{lab} = 1.5 - 15$ GeV/c)

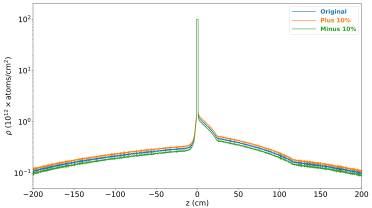


Restgas density distribution along the z-axis

Reality in the Experiment

Sensitivity study

- Vary the Restgas density by $\pm 10\%$.
- Consistent model: luminosity deviation remains negligible.
- Mismatched model: bias reaches $\sim \pm 3.4\%$.



Target Profiles	$\Delta L/L$ [%] Consistent Model	$\Delta L/L$ [%] Original Model
-10% Restgas	-0.02 ± 0.08	-3.39 ± 0.08
Original	0.02 ± 0.08	0.02 ± 0.08
+10% Restgas	0.02 ± 0.08	3.39 ± 0.08

High precision in luminosity determination requires a precise Restgas profile.

Reconstruction Bottleneck

Why standard reconstruction fails for Restgas background?

- Beam-Restgas interactions distributed along the entire beam pipe.
- Standard Track finder (STTMVDGEM/BARREL) optimized for IP at $(0, 0, 0)$.
- PID step back-propagated tracks to the z-axis for vertex determination.

Consequences of Displaced Tracks

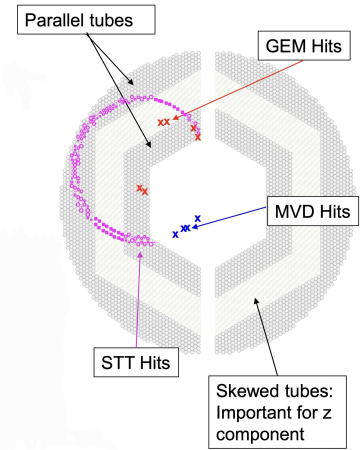
- Wrong reconstructed vertex.
- Bias in dE/dx and momentum vector measurements.
- Low reconstruction efficiency.

Track Finder for Displaced Vertices

Apollonius Triplet Track Finder - A Tailored Track Finder for Secondaries

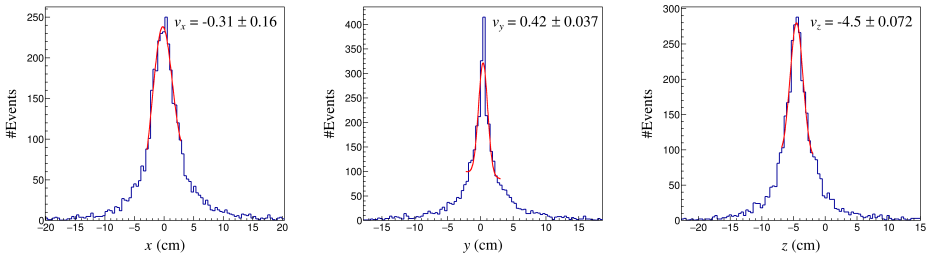
1. Apollonius circles in xy-plane
   - Select STT hits
   - Calculate circles
   - Chose a proper subset circle
2. Longitudinal reconstruction in z-axis
   - Pz Finder with skewed STT hits
   - Calculate z position and p_z
3. Global Tracking
   - KalmanFilter with MVD & GEM hits.

Developed by Anna Alicke, JÜLICH



A POCA-First Approach:

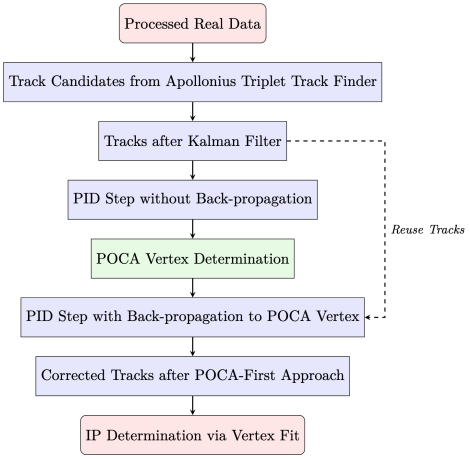
1. PID step without back-propagation.
2. Vertex determination via POCA of all $\bar{p}p$ tracks.

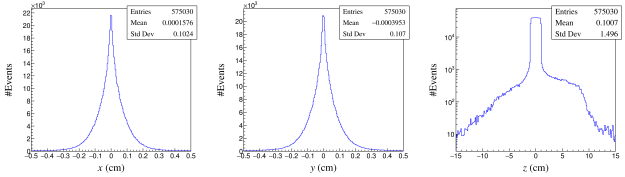


simulated events with a true vertex at $x = -0.2\,\mathrm{cm},\ y = 0.4\,\mathrm{cm},\ z = -5.0\,\mathrm{cm}$

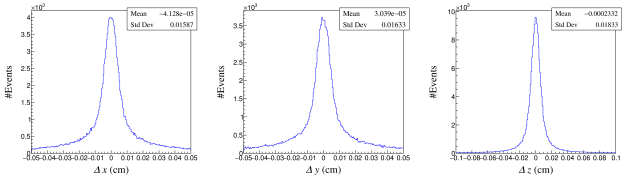
3. PID step with back-propagation to the POCA vertex (*POCA-to-point*).

Measurement of the Restgas Distribution



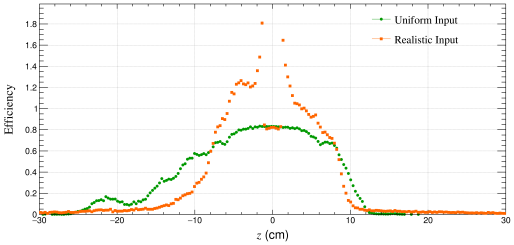


Reconstructed Restgas distribution

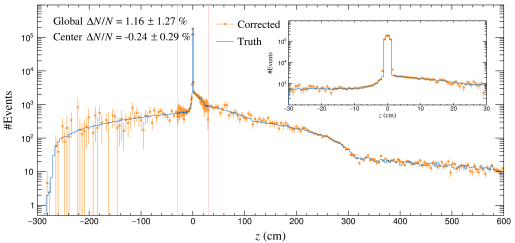


Resolution $(x, y, z)_{\mathrm{reco}} - (x, y, z)_{\mathrm{mc}}$

Efficiency Correction



Profile-Dependent Efficiency



Baseline Validation

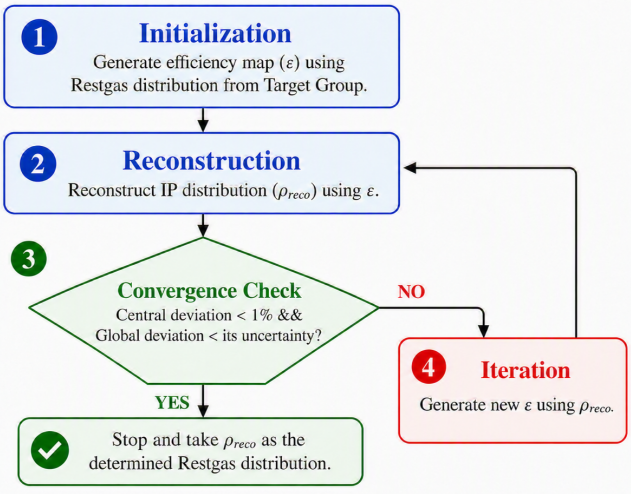
Forward Modeling Approach

- Generate MC samples using a realistic input density profile $\rho_{\mathrm{in}}(z)$.
- Calculate efficiency $\epsilon = N_{\mathrm{reco}}/N_{\mathrm{gen}}$.

Circular Dependency

- Restgas distribution needed for Restgas determination.

Iterative Workflow



Iteration Results

- An example of iteration starting with 40% overestimation of the Restgas distribution.

Iterations	Global [%]	Central [%]
1	12.27 ± 5.69	1.58 ± 0.30
2	4.28 ± 1.19	1.18 ± 0.28
3	1.30 ± 1.32	-0.22 ± 0.29

The iteration converges rapidly!

Luminosity Determination

- $p_{\text{beam}} = 4.06$ GeV/c.

Target Profiles	Fit Model	$\Delta L/L$ [%]
Point-like Target	Point-like Target	-0.04 ± 0.07
Target with Restgas	MC Target Distribution	0.02 ± 0.08
Target with Restgas	Measured Target Distribution	0.05 ± 0.07

Summary

- LuminosityFit extended to include Restgas background and achieved $|\Delta L/L| < 0.5\%$ for $p_{\mathrm{beam}} = 1.5 - 15\,\mathrm{GeV}/c$.
- Algorithms were developed to determine the Restgas distribution with high precision.
- Implemented the measured Restgas distribution, Luminosity could be determined with uncertainty below 0.5%.

The precision of Luminosity Determination could achieve the requirement of PANDA physics program.



Thanks for your attention!